# SEAT BELT BUCKLE REPLACEMENT

**AFFECTED VEHICLES:** **CERTAIN 2001 MODEL YEAR TAURUS AND SABLE**

## OVERVIEW

This procedure details the replacement of driver and passenger seat belt buckles on the affected vehicles. It also includes a functionality test for the ***new*** seat belt buckles.

## SERVICE PROCEDURE

***WARNING!***

***ALWAYS WEAR SAFETY GLASSES WHEN REPAIRING AN AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) VEHICLE AND WHEN HANDLING AN AIRBAG MODULE. THIS WILL REDUCE THE RISK OF INJURY IN THE EVENT OF ACCIDENTAL DEPLOYMENT.***

***WARNING!***

***AFTER DEPLOYMENT, THE AIR BAG SURFACE CAN CONTAIN DEPOSITS OF SODIUM HYDROXIDE, A BYPRODUCT OF THE GAS GENERATED DURING COMBUSTION, THAT IS IRRITATING TO THE SKIN. WASH YOUR HANDS WITH SOAP AND WATER IMMEDIATELY AFTER HANDLING DEPLOYED AIR BAGS.***

***WARNING!***

***NEVER PROBE THE CONNECTORS ON THE AIR BAG MODULE. DOING SO CAN RESULT IN AIR BAG DEPLOYMENT, WHICH CAN RESULT IN PERSONAL INJURY.***

***WARNING!***

***VEHICLE SENSOR ORIENTATION IS CRITICAL FOR PROPER SYSTEM OPERATION. IF A VEHICLE EQUIPPED WITH AN AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) IS INVOLVED IN A COLLISION, INSPECT THE SENSOR MOUNTING BRACKET AND WIRING PIGTAIL FOR DEFORMATION. REPLACE AND PROPERLY POSITION THE SENSOR OR ANY OTHER DAMAGED SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS WHETHER OR NOT THE AIR BAG IS DEPLOYED.***

***WARNING!***

***THE RESTRAINT SYSTEM DIAGNOSTIC TOOL IS FOR RESTRAINT SYSTEM SERVICE ONLY. REMOVE FROM VEHICLE PRIOR TO ROAD USE. FAILURE TO REMOVE COULD RESULT IN INJURY AND POSSIBLE VIOLATION OF VEHICLE SAFETY STANDARDS.***

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01

NOTE: To perform this procedure, you will need the following special tools:

- Restraint System Diagnostic Tool 418-F403 (2 required).
- Restraint System Diagnostic Tool 418-133 (2 required).
- Restraint System Diagnostic Tool 418-F405.
- Restraint System Diagnostic Tool 418-F407.

1. Move and tilt both front seats to their highest and rearmost position.
2. **WARNING: TO AVOID ACCIDENTAL AIR BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY FRONT OR SIDE AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS AND BEFORE SERVICING, REPLACING, ADJUSTING OR STRIKING COMPONENTS NEAR THE FRONT OR SIDE AIR BAG SENSORS, SUCH AS DOORS, INSTRUMENT PANEL, CONSOLE, DOOR LATCHES, STRIKERS, SEATS AND HOOD LATCHES.**

   Disconnect the battery negative cable and wait at least two (2) minutes. Be sure to disconnect auxiliary batteries and power supplies (if equipped).
3. Remove the lower steering column opening cover with reinforcement.
   - Remove the bolts.
   - Pull out to release the retaining clips and remove the lower steering column opening cover with reinforcement.
4. Disconnect the four-pin clockspring electrical connector at the base of the steering column. Attach Restraint System Diagnostic Tool 418-F403 to the vehicle harness side of the four-pin clockspring electrical connector.
5. Open the glove box to its full extent by pushing in on the side tabs, release the glove box.
6. Through the glove box opening, remove the passenger front air bag module retaining bolts.
7. Remove the passenger front air bag module and trim cover.
   - Using a typical trim tool, pull the left-hand corner of the trim cover away from the instrument panel.
   - From left to right, slide the trim tool across the seam between the trim cover and instrument panel, releasing the trim cover retaining clips.
   - Pull the passenger front air bag module and trim cover away from the instrument panel, but do not remove it.
   - Disconnect the passenger front air bag module electrical connector.
   - Remove the wire harness pin-type retainer from the passenger front air bag module.
   - Remove the passenger front air bag module.
8. Attach Restraint System Diagnostic Tool 418-F403 to the vehicle harness side of the passenger front air bag module electrical connector.
9. NOTE: In steps 9 and 10, the connectors referred to are located under their respective seats, on the outboard sides.

   Disconnect the passenger and driver side air bag module electrical connectors, if equipped, and attach Restraint System Diagnostic Tool 418-133 to the vehicle harness side of the electrical connectors.

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01

10. NOTE: The restraint system diagnostic tools for the driver and passenger safety belt pretensioners are similar but not interchangeable. Make sure to install the correct tool in the correct connector.

    Disconnect the driver and passenger seat 10-pin electrical connectors. Then attach Restraint System Diagnostic Tool 418-F405 to the vehicle harness side of the driver 10-pin connector and Tool 418-F407 to the vehicle harness side of the passenger 10-pin connector.

11. Remove the driver and passenger seat front trim covers, then remove the bolts securing the front of the driver and passenger seats to the vehicle floor.

12. If necessary for step 13, reconnect the battery negative cable.

13. Move both seats forward.

14. **WARNING: TO AVOID ACCIDENTAL AIR BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY FRONT OR SIDE AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS AND BEFORE SERVICING, REPLACING, ADJUSTING OR STRIKING COMPONENTS NEAR THE FRONT OR SIDE AIR BAG SENSORS, SUCH AS DOORS, INSTRUMENT PANEL, CONSOLE, DOOR LATCHES, STRIKERS, SEATS AND HOOD LATCHES.**

    Disconnect the battery negative cable, if connected in step 12, and wait at least two (2) minutes. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

15. Remove the driver and passenger seat rear trim covers, then remove the nuts securing the rear of the driver and passenger seats to the vehicle floor.

16. Remove the driver and passenger seats from the vehicle and place both seats on a clean work area for replacement of the seat belt buckles.

17. Release the seat cushion trim cover side J-clips.

18. Disconnect and route the safety belt buckle and pretensioner electrical connector and wire harness out from under the seat.

19. **WARNING: DO NOT HANDLE THE SAFETY BELT BUCKLE AND PRETENSIONER IN THE AREA OF THE SAFETY BELT BUCKLE AND PRETENSIONER CABLE AND RETAINING NUT. IF ACCIDENTAL PRETENSIONER DEPLOYMENT OCCURS, IT CAN RESULT IN PERSONAL INJURY.**

    Remove the safety belt buckle and pretensioner assembly from the seat assembly.

20. Position the ***new*** safety belt buckle and pretensioner assembly onto the seat assembly.

21. Install the fastener and tighten it to 36 Nm (27 lb-ft).

22. Route the safety belt buckle and pretensioner electrical connector and wire harness between the seat cushion frame and the seat track. Then connect the safety belt buckle and pretensioner electrical connector.

23. Reposition the seat cushion trim cover over the safety belt buckle and pretensioner and attach the J-clips.

24. Reconnect the battery negative cable.

25. Prove out the supplemental restraint system. For additional information, or if prove-out fails, refer to Section 501-20B: Occupant Restraint — Passive — Supplemental Air Bag System: Diagnosis and Testing in the 2001 Taurus/Sable Workshop Manual.

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01

26. **WARNING: TO AVOID ACCIDENTAL AIR BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY FRONT OR SIDE AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS AND BEFORE SERVICING, REPLACING, ADJUSTING OR STRIKING COMPONENTS NEAR THE FRONT OR SIDE AIR BAG SENSORS, SUCH AS DOORS, INSTRUMENT PANEL, CONSOLE, DOOR LATCHES, STRIKERS, SEATS AND HOOD LATCHES.**

    Disconnect the battery negative cable and wait at least two (2) minutes. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

27. Remove the restraint system diagnostic tool from the vehicle harness side of the driver and passenger seat 10-pin electrical connectors.

28. Position both seats into the vehicle.

29. Tilt the driver seat toward the rear of the vehicle. Connect the driver seat 10-pin electrical connector. Repeat for the passenger side.

30. Install the nuts securing the rear of the driver and passenger seats to the vehicle floor. Tighten inboard nuts to 80 Nm (59 lb-ft) and outboard nuts to 30 Nm (22 lb-ft), then install the driver and passenger seat rear trim covers.

31. If necessary for step 32, reconnect the battery negative cable.

32. Move and tilt both front seats to their highest and rearmost position.

33. **WARNING: TO AVOID ACCIDENTAL AIR BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY FRONT OR SIDE AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS AND BEFORE SERVICING, REPLACING, ADJUSTING OR STRIKING COMPONENTS NEAR THE FRONT OR SIDE AIR BAG SENSORS, SUCH AS DOORS, INSTRUMENT PANEL, CONSOLE, DOOR LATCHES, STRIKERS, SEATS AND HOOD LATCHES.**

    Disconnect the battery negative cable, if connected in step 31, and wait at least two (2) minutes. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

34. Install the bolts securing the front of the driver and passenger seats to the vehicle floor. Tighten all bolts to 30 Nm (22 lb-ft), then install the driver and passenger seat front trim covers.

35. Remove the restraint system diagnostic tool from the vehicle harness side of the driver and passenger side air bag module electrical connectors, if equipped. Connect the driver and passenger side air bag module electrical connectors.

36. Reconnect the battery negative cable.

37. NOTE: Restraint System Diagnostic Tools 418-F407 and 418-F405 may cause supplemental restraint system (SRS) diagnostic trouble codes (DTC). Because of this, you must connect the NGS or WDS to the vehicle and clear any SRS DTCs before proceeding with the prove-out procedure in this step.

    With the restraint system diagnostic tools still connected to the driver and passenger front airbag connectors, prove out the supplemental restraint system (SRS). For additional information, refer to Section 501-20B: Occupant Restraint — Passive — Supplemental Air Bag System: Diagnosis and Testing in the 2001 Taurus/Sable Workshop Manual.

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01

38. **WARNING: TO AVOID ACCIDENTAL AIR BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY FRONT OR SIDE AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS AND BEFORE SERVICING, REPLACING, ADJUSTING OR STRIKING COMPONENTS NEAR THE FRONT OR SIDE AIR BAG SENSORS, SUCH AS DOORS, INSTRUMENT PANEL, CONSOLE, DOOR LATCHES, STRIKERS, SEATS AND HOOD LATCHES.**

    Disconnect the battery negative cable and wait at least two (2) minutes. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

39. Remove the restraint system diagnostic tool from the vehicle harness side of the passenger front air bag module electrical connector.

40. NOTE: Make sure there are no foreign objects in the passenger front air bag module alignment channels.

    Remove or position out of the way anything that is blocking the accessibility to the instrument panel alignment rails or holes.

    Install the passenger front air bag module and trim cover into the instrument panel.
    - Connect the passenger front air bag module electrical connector.
    - Install the wire harness pin-type retainer to the passenger front air bag module.
    - Position the passenger front air bag module alignment channels to the alignment rails of the instrument panel.
    - Starting at the left-hand side of the trim cover, install the upper and lower alignment pins into the instrument panel.
    - Working from left to right, install all the trim cover retaining tabs and alignment pins into the instrument panel.
    - When all channels and tabs are aligned correctly, there will be an even gap around the perimeter of the passenger front air bag trim cover.

41. Through the glove box opening, install the passenger front air bag module retaining bolts. Tighten the bolts to 8 Nm (71 lb-in) and close the glove box.

42. Remove the restraint system diagnostic tool from the vehicle harness side of the four-pin clockspring electrical connector at the base of the steering column. Connect the four-pin clockspring electrical connector to the vehicle harness connector.

43. Install the lower steering column opening cover with reinforcement.
    - Position the lower steering column opening cover with reinforcement to the instrument panel and push in at the top, seating the retaining clips.
    - Install the bolts. Tighten the bolts to 7 Nm (62 lb-in).

44. Reconnect the battery negative cable.

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01

45. **WARNING: THE RESTRAINT SYSTEM DIAGNOSTIC TOOLS ARE FOR RESTRAINT SYSTEM SERVICE ONLY. REMOVE THEM FROM THE VEHICLE PRIOR TO ROAD USE. FAILURE TO REMOVE CAN RESULT IN INJURY AND POSSIBLE VIOLATION OF VEHICLE SAFETY STANDARDS.**

    With all the restraint system diagnostic tools removed, prove out the supplemental restraint system (SRS). For additional information, refer to Section 501-20B: Occupant Restraint — Passive — Supplemental Air Bag System: Diagnosis and Testing in the 2001 Taurus/Sable Workshop Manual.

46. Perform a Component Functionality Test on the ***new*** seat belt buckles as follows:

    NOTE: Since both seat belt buckles have been replaced, an assistant must be used in the passenger seat.

    - Fasten safety belts and proceed to a safe area.
    - Attain a speed of 8 km/h (5 mph).
    - **WARNING: DRIVER AND PASSENGER MUST BE PREPARED TO BRACE THEMSELVES IF THE RETRACTOR DOES NOT LOCK.**

      Grasp the shoulder harness and prepare to lean forward.
    - Make a maximum brake application without a skid.
    - NOTE: Do not jerk on the safety belt webbing while carrying out this test.

      Lean forward slightly when the brake application is made. The safety belts should lock up with minimum webbing extension.
    - If both shoulder straps lock up, the safety belt assemblies are functioning correctly. If either or both retractors fail to lock up at the 8km/h (5 mph) speed, repeat the test at a constant 24 km/h (15 mph) speed.
    - NOTE: If either or both shoulder belts do not lock up, the sheet metal in the retractor's mounting surface may need to be returned to its original shape and structural integrity. If either or both shoulder belts do not lock up at the 24 km/h (15 mph) test, return the vehicle for repair of the safety belts.

Ford Motor Company
CPR © 2001 FORD MOTOR COMPANY
DEARBORN, MICHIGAN 48121
04/01